FODS ASSIGNMENT 5

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Batch – 2	
DATE – 11-08-2026	TIME – 8:00 TO 10:00

Question - Practical 5: Searching Algorithms

Implement Linear Search, Sentinel Search, Binary Search, Fibonacci Search, and Indexed Sequential Search. Compare their performance for different input sizes.

Answer :

CODE :

1. **Implement Linear Search :**

```
#include <iostream>
using namespace std;

int main(){
   int n , key;
   cout<<"Enter no. of elements:";
   cin>>n;

   int array[n];

   cout<<"Enter the elements:";
   for(int i=0;i<n;i++){
      cin>>array[i];
   }

   cout<<"Enter the element to search:";
   cin>>key;

   for(int i=0;i<n;i++){
     if (array[i]==key)
     {
      cout<<"Element found at position:" << i <<endl;
      return 0;
```

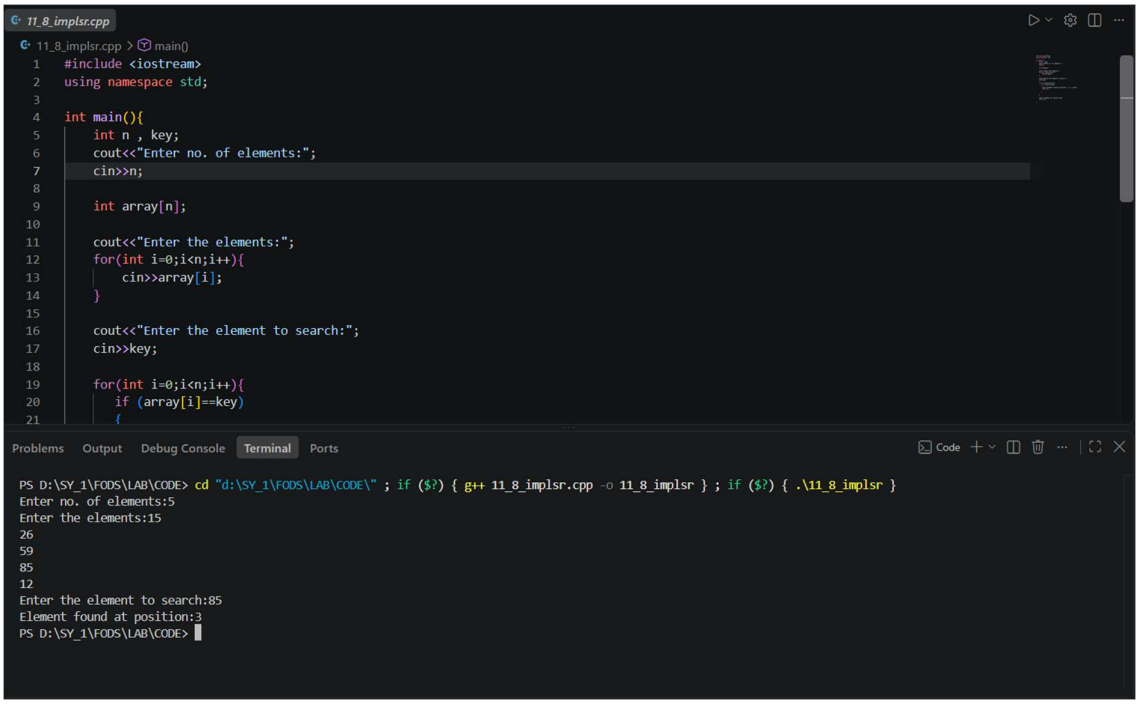
```
    }

  }

  cout<<"element not found"<<endl;
  return 0;

}
```

OUTPUT:



2. **Sentinel Search :**

CODE :

```
#include <iostream>
using namespace std;
int main() {
  int n, key;

  cout << "Enter number of elements: ";
  cin >> n;

  int arr[n + 1];

  cout << "Enter elements: ";
  for (int i = 0; i < n; i++) {
    cin >> arr[i];
  }
```

```
    cout << "Enter element to search: ";
    cin >> key;


    arr[n] = key;

    int i = 0;

    while (arr[i] != key) {
        i++;
    }

    if (i < n)
        cout << "Element found at position " << i + 1 << endl;
    else
        cout << "Element not found" << endl;

    return 0;
}
```

OUTPUT:



3. **Binary Search:**

CODE:

```
#include <iostream>
using namespace std;
```

```
int main() {
   int n, key;
   cout << "Enter number of elements: ";
   cin >> n;

   int arr[n];

cout << "Enter elements in sorted order: ";

   for (int i = 0; i < n; i++) {
      cin >> arr[i];
   }

   cout << "Enter element to search: ";
   cin >> key;

   int low = 0;
   int high = n - 1;
   int pos = -1;

   while (low <= high) {
      int mid = low + (high - low) / 2;

      if (arr[mid] == key) {
         pos = mid;
         break;
      }
      else if (arr[mid] < key) {
         low = mid + 1;
      }
      else {
         high = mid - 1;
      }
   }

   if (pos != -1)
      cout << "Element found at position " << pos + 1 << endl;
   else
      cout << "Element not found" << endl;

   return 0;
}
```

OUTPUT:



4. **Fibonacci Search :**

CODE:

```
#include <iostream>
using namespace std;

int main() {
   int n, key;

   cout << "Enter number of elements: ";
   cin >> n;

   int arr[n];

   cout << "Enter elements in sorted order: ";
   for (int i = 0; i < n; i++) {
      cin >> arr[i];
   }

   cout << "Enter element to search: ";
   cin >> key;

   int fib2 = 0;
   int fib1 = 1;
   int fib = fib1 + fib2;


   while (fib < n) {
      fib2 = fib1;
```

```
    fib1 = fib;
    fib = fib1 + fib2;
  }

  int offset = -1;

  while (fib > 1) {
    int i = min(offset + fib2, n - 1);

    if (arr[i] < key) {
      fib = fib1;
      fib1 = fib2;
      fib2 = fib - fib1;
      offset = i;
    }
    else if (arr[i] > key) {
      fib = fib2;
      fib1 = fib1 - fib2;
      fib2 = fib - fib1;
    }
    else {
      cout << "Element found at position " << i + 1 << endl;
      return 0;
    }
  }

  if (fib1 && offset + 1 < n && arr[offset + 1] == key) {
    cout << "Element found at position " << offset + 2 << endl;
  }
  else {
    cout << "Element not found" << endl;
  }

  return 0;
}
```

OUTPUT:



5. **Indexed Sequential Search :**

CODE:

```
#include <iostream>
#include <cmath>
using namespace std;

int main() {
   int n, key;

   cout << "Enter number of elements: ";
   cin >> n;

   int arr[n];

   cout << "Enter elements in sorted order: ";
   for (int i = 0; i < n; i++) {
      cin >> arr[i];
   }

   cout << "Enter element to search: ";
   cin >> key;

   int blockSize = sqrt(n);

   int start = 0;
   int end = blockSize - 1;
```

```
    while (end < n && arr[end] < key) {
        start = end + 1;
        end = end + blockSize;
    }

    if (end >= n)
        end = n - 1;

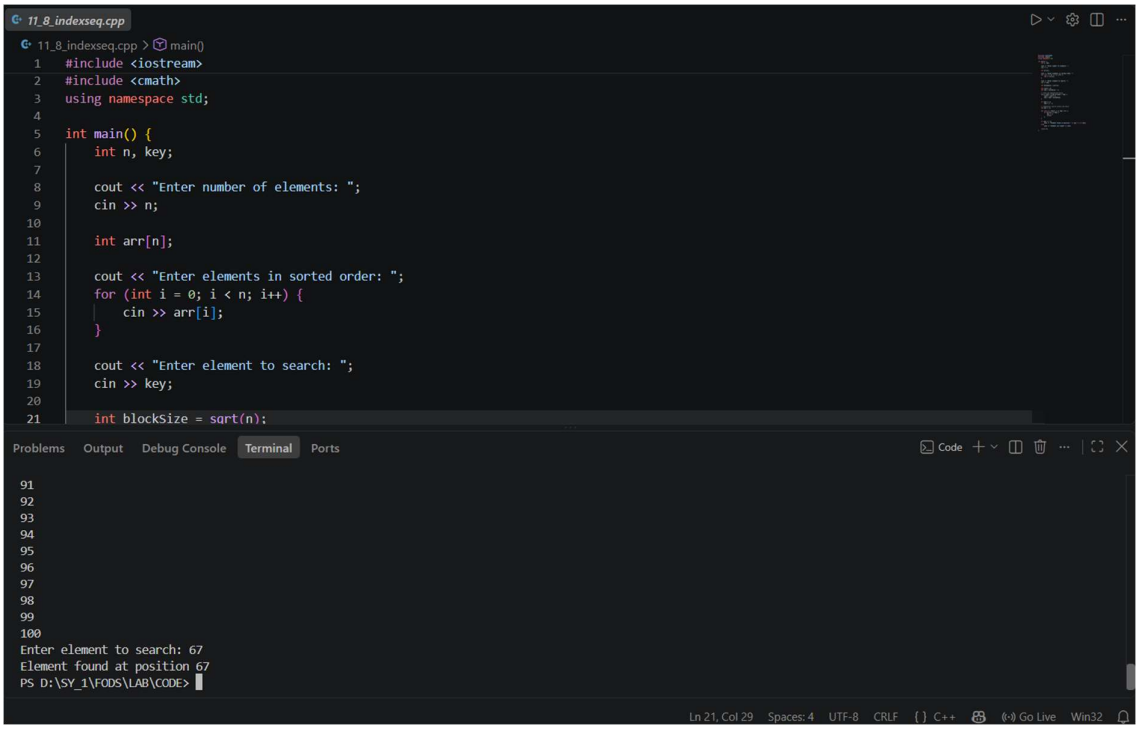
    // Sequential search within the block
    int pos = -1;

    for (int i = start; i <= end; i++) {
        if (arr[i] == key) {
            pos = i;
            break;
        }
    }

    if (pos != -1)
        cout << "Element found at position " << pos + 1 << endl;
    else
        cout << "Element not found" << endl;

    return 0;
}
```

OUTPUT:



- **DIFFERENCE:**

Algorithm	**Worst-Case**	**Average-Case**	**Best-Case**	**Algorithm**	**Worst-Case**
Fibonacci Search	O(log n)	O(log n)	O(1)	Fibonacci Search	O(log n)
Linear Search	O(n)	O(n)	O(1)	Linear Search	O(n)
Sentinel Search	O(n)	O(n)	O(1)	Sentinel Search	O(n)